

HOMEWORK 7
ELEMENTARY LINEAR ALGEBRA
(MATH 2250-03), SPRING, 2018

CHECK: 04/12/2019

SECTION 4.5 - THE DIMENSION OF A VECTOR SPACES

1. Let H be a nonzero subspace of a finite-dimensional vector space V and let $T : V \rightarrow W$ be a linear transformation. Show that $T(H) = \{T(h) : h \in H\}$ is a subspace of W and prove that $\dim(T(H)) \leq \dim(H)$.
2. Let $M_{n \times n}(\mathbb{R})$ be the vector space of all $n \times n$ matrices with real entries, and let $Sym_n(\mathbb{R})$ denote the subset of $n \times n$ matrices which are equal to their transpose, i.e.

$$Sym_n(\mathbb{R}) = \{A \in M_{n \times n}(\mathbb{R}) : A = A^T\}.$$

Such matrices are generally called *symmetric matrices*. Show that $Sym_n(\mathbb{R})$ is a vector space and compute $\dim(Sym_n(\mathbb{R}))$.

SECTION 4.6 - RANK

3. Let $A = \begin{bmatrix} 1 & -4 & 9 & -7 \\ -1 & 2 & -4 & 1 \\ 5 & -6 & 10 & 7 \end{bmatrix}$. The matrix A is row-equivalent to the matrix $\begin{bmatrix} 1 & 0 & -1 & 5 \\ 0 & -2 & 5 & -6 \\ 0 & 0 & 0 & 0 \end{bmatrix}$.

Without calculations, compute $\text{rank}(A)$ and $\dim(\text{Nul}(A))$. Then, find bases for $\text{Col}(A)$, $\text{Row}(A)$, and $\text{Nul}(A)$.

4. Suppose A is an $m \times n$ matrix with real entries and suppose $b \in \mathbb{R}^n$. What must be true of $\text{rank}\left(\begin{bmatrix} A & b \end{bmatrix}\right)$ and $\text{rank}(A)$ in order for the equation $Ax = b$ to be consistent?
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SECTION 4.7 - CHANGE OF BASIS

5. In parts (a) and (b) of this exercise, you may assume that $\mathcal{B} = \{b_1, b_2\}$ and $\mathcal{D} = \{d_1, d_2\}$ are bases for $\mathbb{R}^2$. Find the change-of-coordinates matrix from $\mathcal{B}$ to $\mathcal{D}$.

(a) $b_1 = \begin{bmatrix} 7 \\ 5 \end{bmatrix}, b_2 = \begin{bmatrix} -3 \\ -1 \end{bmatrix}, d_1 = \begin{bmatrix} 1 \\ -5 \end{bmatrix}, d_2 = \begin{bmatrix} -2 \\ 2 \end{bmatrix}$

(b) $b_1 = \begin{bmatrix} -6 \\ -1 \end{bmatrix}, b_2 = \begin{bmatrix} 2 \\ 0 \end{bmatrix}, d_1 = \begin{bmatrix} 2 \\ -1 \end{bmatrix}, d_2 = \begin{bmatrix} 6 \\ -2 \end{bmatrix}$

6. In $\mathbb{P}_2$, find the change-of-coordinates matrix from the basis $\mathcal{B} = \{1 - 2t + t^2, 3 - 5t + 4t^2, 2t + 3t^2\}$ to the standard basis $\mathcal{C} = \{1, t, t^2\}$. Then, using this matrix, find the $\mathcal{B}$ -coordinate vector $[p]_{\mathcal{B}}$ for the polynomial $p(t) = -1 + 2t$.